Graph Convolutional Networks with Adaptive Gating for Channel Selection in Motor Imagery BCI

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Abstract

Background and Objective: Motor imagery brain-computer interfaces require efficient channel selection to obtain high classification while reducing hardware complexity and enhancing system usability. This paper proposes an end-to-end deep learning system that is an adaptive gating method that utilizes graph convolutional networks to learn the relative importance of EEG channels.

Methods: The proposed architecture integrates both temporal feature extraction and graph-based spatial reasoning as well as uses learnable adaptive gating to control the individual channel contribution. According to the parameters learned, three strategies of channel selection namely edge-selection, aggregation-selection and gate-selection. The model is tested on four datasets that have 22 to 118 channels using 5-fold cross-validation.

Results: Adaptive gating had an accuracy improvement of 1.1–3.7% over baseline with accuracy improvements of 83.3% (PhysioNet), 85.3% (BCI IV IIa), 80.0% (BCI III IVa) and 88.9% (BCI III IIIa). The gate-based selection retained 99.6% accuracy with 68.75% channel reduction (64 to 20), which was better than edge-selection (96.1%) and aggregation-selection (95.8%).

Conclusions: Adaptive gating allows a good reduction of channels without

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compromising on accuracy. The chosen channels are compatible with sensorimotor areas (C3, C4, Cz, CPz), which allows convenient use of BCI with less expensive hardware and enhanced comfort levels of users.
Keywords: Brain-computer interface, Motor imagery, EEG channel selection, Graph convolutional network, Adaptive gating, Deep learning

1. Introduction

Motor imagery (MI) brain-computer interfaces can be used to communicate and control using the imagined movements, and they are applied to neuro-rehabilitation and assistive technology [1, 2, 3, 4, 5, 6]. Nevertheless, there are two concerns that restrict the useful application: (1) classification performance is still suboptimal, and (2) dense electrode arrays (64-118 channels) are more expensive for hardware, uncomfortable for the user and computationally intensive. The multichannel EEG signals have both opportunities and challenges because spatial information plays a dominant role in differentiating the motor imagery patterns, but too many channels would lead to redundancy and noise.

The Traditional methods incorporate the CNNs [7, 8, 9, 10] to extract the temporal features, which fail to address the discrete spatial nature of the EEG. The non-Euclidean geometry of the scalp generated by the irregular positioning of electrodes cannot be captured by CNNs. Graph neural networks (GNNs) [11, 12, 13, 14, 15] overcome this, but the explicit relationship model of electrode relationships with learned adjacency matrices, but most approaches instead present manually designed adjacency matrices with respect to the distance between electrodes or domain knowledge [16, 17, 18]. This dependence on prior knowledge limits the possibility of being adaptable to subject-specific brain processes.

The methods of channel selection are intended to discover electrodes that are relevant to the task, without affecting performance. Common Spatial Patterns (CSP) [19, 20, 21] and its variants [22, 23, 24] are algorithms that utilize handcrafted features but do not consider spatial relationships among channels. Wrapper methods cannot scale to high-dimensional EEG data. More recent deep learning techniques [25, 26, 27, 28, 29], such as attention maps [30] or squeeze-and-excitation blocks [31], make automated decisions but cannot be neurophysiologically understood and spatially modeled. The discrepancy between theory and practice in classifications inspires the desire to find interpretable end-to-end channel selection which does not violate neurophysiological limits.

This paper suggests adaptive gating strategies of end-to-end channel selection in graph-based MI-BCI classification. Channel importance is learned by the gating mechanism, which learns channel importance alongside classification, allowing interpretable trial-specific modulation. Its contributions include: (1) learnable hybrid gating between static importance and trial-specific modulation, (2) channel-selection technique(gate-based) based on the learned parameters, (3) 99.6% retention of accuracy and 68.75% reduction in the number of channels on four public datasets (22-118 channels), (4) neurophysiologically-valid selections using the learned parameters which are correlated with sensorimotor areas (C3, C4, Cz, CPz).

2. Methods

This paper extends the EEG-ARNN framework that was suggested by Sun et al. [32] and designed adaptive gating schemes, under which we can effectively select channels. The model has two main segments, namely, the temporal feature extraction module (TFEM) and the channel active reasoning module (CARM), as shown in Figure 1. In this section, we show the base architecture, proposed gating mechanisms and channel selection mechanisms.

2.1. Baseline EEG-ARNN Architecture

Multichannel input EEG time-series data $X \in \mathbb{R}^{B \times C \times T}$ is processed by the baseline EEG-ARNN model, where B is batch size, C is the number of channels (electrodes), and T is the number of time points. This architecture consists of alternating temporal convolutions and graph convolution layers.

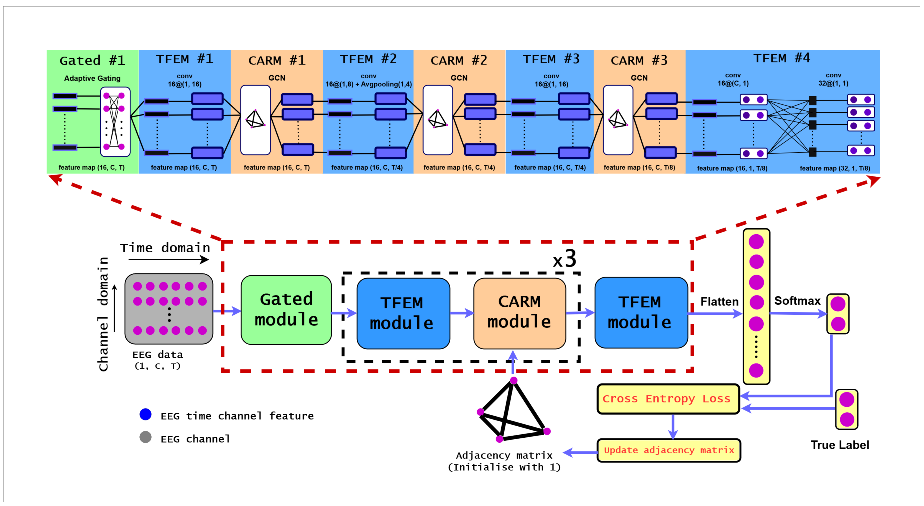

Figure 1: Adaptive gating architecture for Gated EEG-ARNN.

2.1.1. Temporal Feature Extraction Module (TFEM)

Temporal convolution assists in deriving features of the time domain of the EEG signals. Each TFEM layer applies a 1D convolution on the temporal dimension:

$$X^{(l+1)} = \mathrm{ELU}(\mathrm{BN}(\mathrm{Conv1D}(X^{(l)}))) \tag{1}$$

Where Conv1D denotes 1D convolution along the temporal dimension with a kernel size of 16, BN is batch normalization and ELU is the exponential linear unit activation function. Temporal convolution is then succeeded by average pooling with stride of 2 to reduce the time dimension and obtain multi-scale temporal patterns.

2.1.2. Channel Active Reasoning Module (CARM)

The CARM layer adds graph convolutional layers to the model, which helps in understanding the spatial relation between EEG electrodes. The classical GCN methods are grounded on the fixed adjacency matrix from the positions of electrodes [33] which is inconvenient since it has to be done by professionals. Here, the adjacency matrix $A \in \mathbb{R}^{C\times C}$ is trained from the data. A matrix constructed assists in identifying the functional relationship of channels, hence it reflects the real connection more effectively. The adjacency matrix to be learned is initialized with static values and is refurbished during the training:

$$A = \text{sigmoid}(W_A) \tag{2}$$

where $W_A \in \mathbb{R}^{C\times C}$ is a parameter matrix which is learnable for taking symmetry and self-connection in consideration, we compute:

$$\hat{A} = \frac{1}{2}(A + A^T) + I \tag{3}$$

here I is the identity matrix. This is then normalized in the adjacency matrix by the use of symmetric normalization:

$$\tilde{A} = D^{-1/2}\hat{A}D^{-1/2} \tag{4}$$

where D is the degree matrix with $D_{ii} = \sum_j \hat{A}_{ij}$.

Graph convolution is performed by:

$$X^{(l+1)} = \text{ELU}(\text{BN}(\tilde{A}X^{(l)}\Theta^{(l)})) \tag{5}$$

where $\Theta^{(l)}$ is a weight matrix of layer l that is learnable.

The CARM module uses the learnable graph convolution based on spectral graph theory [11, 12]. Following the approach of Kipf and Welling, we use a first-order approximation of spectral filters with renormalization:

$$H_t = \hat{W}\, X_t\, \Theta_t, \quad \text{where} \quad \hat{W} = \tilde{D}^{-\frac{1}{2}}\, \tilde{W}\, \tilde{D}^{-\frac{1}{2}}, \tag{6}$$

with $\tilde{W} = W + I_C$ (self-loops added) and $\tilde{D}_{ii} = \sum_j \tilde{W}_{ij}$. The adjacency matrix W^* is initialized fully-connected and learned end-to-end via gradient descent:

$$W^*_{ij} = \begin{cases} 1, & i \neq j \\ 0, & i = j, \end{cases} \quad ; \quad \hat{W}^* \leftarrow (1-\rho)\,\hat{W}^* - \rho\,\frac{\partial \mathcal{L}}{\partial \hat{W}^*}, \tag{7}$$

where $\rho = 0.001$ is a small learning rate. This adaptive adjacency learning captures subject-specific functional connectivity without requiring predefined electrode distance matrices.

The entire baseline architecture will be represented by three TFEM-CARM blocks with three layers, where the layers are going to be fully connected and categorized:

$$X^{(1)} = \mathrm{CARM}_1(\mathrm{TFEM}_1(X)) \tag{8}$$

$$X^{(2)} = \mathrm{CARM}_2(\mathrm{TFEM}_2(X^{(1)})) \tag{9}$$

$$X^{(3)} = \mathrm{CARM}_3(\mathrm{TFEM}_3(X^{(2)})) \tag{10}$$

$$y = \mathrm{FC}_2(\mathrm{Dropout}(\mathrm{ReLU}(\mathrm{FC}_1(\mathrm{Flatten}(X^{(3)}))))) \tag{11}$$

2.2. Gating Mechanism for Channel Selection

To make channel selection more efficient, this paper introduces adaptive gating mechanism in the training, which chooses the contribution of each channel in each training. In this mechanism, channel importance is not directly determined at the end, the model learns and makes use of channel importance in training.

2.2.1. Learnable Adaptive Gating

This learnable adaptive gating technique is a combination of two things. The first is per-channel learnable gates, and the other is adaptive modulation based on input, which is controlled by per-channel learnable gates. For deciding the gate values for channels, a neural network is used, in which the numerical values related to each channel are processed. Specifically, for each trial and each channel, basic statistical information like mean($\mu_c^{(b)}$) and standard deviation($\sigma_c^{(b)}$) is calculated and given to the gating network, based on the behavior of the channel. It gives weightage to the channels dynamically.

$$\mu_c^{(b)} = \frac{1}{T}\sum_t X_{b,c,t}, \quad \sigma_c^{(b)} = \sqrt{\frac{1}{T}\sum_t (X_{b,c,t} - \mu_c^{(b)})^2} \tag{12}$$

Here, b is the batch index, and c is the channel index. For every channel mean, and standard deviations are calculated separately. These two are concatenated, which are shown as $stats = [\mu^{(b)}, \sigma^{(b)}] \in \mathbb{R}^{B\times 2C}$. This was later used as input for the adaptive gating mechanism.

The adaptive modulation is computed by two-layer MLP:

$$g_{\text{adaptive}} = \text{MLP}(stats) = \text{Tanh}(\text{ReLU}(stats W_1 + b_1)W_2 + b_2) \tag{13}$$

Where $W_1 \in \mathbb{R}^{2C\times C}$, $W_2 \in \mathbb{R}^{C\times C}$ are learned weight matrices which are trained and $\mathbf{b}_1 \in \mathbb{R}^C$, $\mathbf{b}_2 \in \mathbb{R}^C$ are bias terms. The output $g_{\text{adaptive}} \in \mathbb{R}^{B\times C}$ gives us input-stat based channel modulation.

Learnable base gates are computed as:

$$\mathbf{g}_{\text{base}} = \sigma(\mathbf{g}_{\text{base\_logits}}) \tag{14}$$

Where $\mathbf{g}_{\text{base\_logits}} \in \mathbb{R}^C$ is a learnable parameter channel wise vector, and σ is the sigmoid function. These base gates encode channel significance learned on all training samples.

Per-channel hybrid selection weights $\boldsymbol{\alpha} \in \mathbb{R}^C$ are learned to balance and find the optimal base and adaptive components:

$$\boldsymbol{\alpha} = \sigma(\boldsymbol{\alpha}_{\text{logits}}) \tag{15}$$

where $\boldsymbol{\alpha}_{\text{logits}} \in \mathbb{R}^C$ are learnable parameters initialized such that $\boldsymbol{\alpha}$ starts near 0.5, which helps the model learn the optimal per-channel logit.

The adaptive modulation is the sum of the learnable base gates with per-channel hybrid weights to the final gate values:

$$\mathbf{g}_c = \sigma(\alpha_c \cdot g_{\text{base},c} + (1 - \alpha_c) \cdot g_{\text{adaptive},c}) \tag{16}$$

for each channel c, where the combination is taken per-channel-wise. The gated input is calculated using:

$$\mathbf{X}_{\text{gated}} = \mathbf{X} \odot \mathbf{g} \tag{17}$$

where $\odot$ denotes element-wise multiplication and $\mathbf{g} \in \mathbb{R}^{B \times 1 \times C \times 1}$.

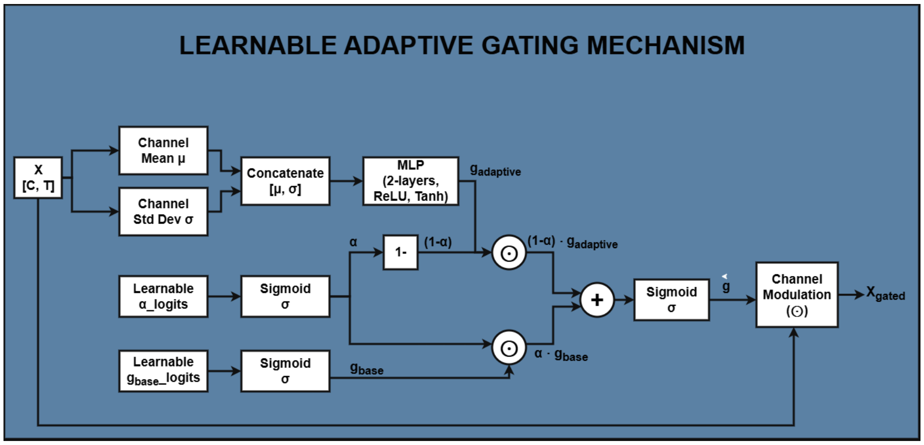

Figure 2: Architecture of the Learnable Adaptive Gating Mechanism. The process calculates per-channel statistics (μ, σ), processes them through a two-layer MLP for adaptive modulation, and combines them with learnable base gates via hybrid weights α to produce the final gate values g for element-wise multiplication $(\odot)$ with the input signal X.

The complete architecture of this mechanism is illustrated in Figure 2. It is a learnable adaptive gating mechanism that dynamically weights channels by trained per-channel importance (base gates) and input-dependent modulation (learnable adaptive gates), and per-channel learnable weights that determine the optimum balance of each electrode.

2.2.2. Training Procedure for Adaptive Gated EEG-ARNN

Algorithm 1 presents the complete training procedure for the Adaptive Gated EEG-ARNN architecture , which uses learnable per channel adaptive gating along with the baseline EEG-ARNN framework.

2.3. Channel Selection Methods

For motor imagery classification, we use three channel selection techniques: edge selection, aggregation selection [32], and gate-based selection (our contribution). The edge selection and aggregation selection methods from Sun et al. [32] directly give importance to channels by the weight matrix obtained by

Algorithm 1 Training Procedure of Adaptive Gated EEG-ARNN

Require: $\mathbf{X} \in \mathbb{R}^{B \times C \times T}$; labels $\mathbf{L}$; adjacency $\hat{\mathbf{W}}^*$; learning rate ρ; epochs n; λ_{gate}
Ensure: Prediction $\mathbf{L}_p$; trained $\hat{\mathbf{W}}^*$; gates $\mathbf{g}$

Initialize: $\mathbf{g}_{\text{base\_logits}}$, $\boldsymbol{\alpha}_{\text{logits}}$, gate network $\{\mathbf{W}_1, \mathbf{b}_1, \mathbf{W}_2, \mathbf{b}_2\}$, $\hat{\mathbf{W}}^*$
for epoch $= 1$ to n **do**
  for each batch $(\mathbf{X}, \mathbf{Y})$ **do**
    Compute $\mu_c^{(b)} = \frac{1}{T}\sum_t X_{b,c,t}$, $\sigma_c^{(b)} = \sqrt{\frac{1}{T}\sum_t (X_{b,c,t} - \mu_c^{(b)})^2}$
    $\textbf{stats} = [\boldsymbol{\mu}^{(b)}, \boldsymbol{\sigma}^{(b)}]$
    $\mathbf{g}_{\text{adaptive}} = \text{Tanh}(\text{ReLU}(\textbf{stats}\mathbf{W}_1 + \mathbf{b}_1)\mathbf{W}_2 + \mathbf{b}_2)$
    $\mathbf{g}_{\text{base}} = \sigma(\mathbf{g}_{\text{base\_logits}})$; $\boldsymbol{\alpha} = \sigma(\boldsymbol{\alpha}_{\text{logits}})$
    $\mathbf{g}_c = \sigma(\alpha_c \cdot g_{\text{base},c} + (1 - \alpha_c) \cdot g_{\text{adaptive},c})\ \forall c$
    $\mathbf{X}_{\text{gated}} = \mathbf{X} \odot \mathbf{g}$; $\mathbf{X}^{(0)} = \mathbf{X}_{\text{gated}}$
    for $k = 1$ to 3 **do**
      $\mathbf{X}^{(k)} = \text{CARM}_k(\text{TFEM}_k(\mathbf{X}^{(k-1)}))$
    end for
    $\mathbf{L}_p = \text{FC}_2(\text{Dropout}(\text{ReLU}(\text{FC}_1(\text{Flatten}(\mathbf{X}^{(3)})))))$
    $\mathcal{L}_{\text{cls}} = \text{CrossEntropy}(\mathbf{L}_p, \mathbf{Y})$; $\mathcal{L}_{\text{gate}} = \lambda_{\text{gate}}\|\mathbf{g}_{\text{base}}\|_1$
    $\mathcal{L} = \mathcal{L}_{\text{cls}} + \mathcal{L}_{\text{gate}}$
    Backpropagate $\nabla_{\boldsymbol{\theta}}\mathcal{L}$
    $\hat{\mathbf{W}}^* \leftarrow (1 - \rho)\hat{\mathbf{W}}^* - \rho\frac{\partial \mathcal{L}}{\partial \hat{\mathbf{W}}^*}$; update others via Adam
  end for
end for

training, whereas gate selection uses the gating parameters to determine their importance. Figure 3 provides a visual overview of how these methods operate.

On all channels, all three techniques produce a scalar score for each electrode after having trained the model on the entire channel set. Based on these importance scores, channels are ranked and the top-K channels are selected (k values vary by dataset). Following selection, the model is retrained with 5-fold cross-validation using only the selected channels, allowing us to determine the accuracy difference between the reduced channel set and the full channel configuration. Detailed experiments across all datasets evaluating classification accuracy, subject-to-subject variability, and spatial consistency demonstrate effective channel reduction capabilities. In several cases, channel selection removes noisy channels, leading to improved performance.

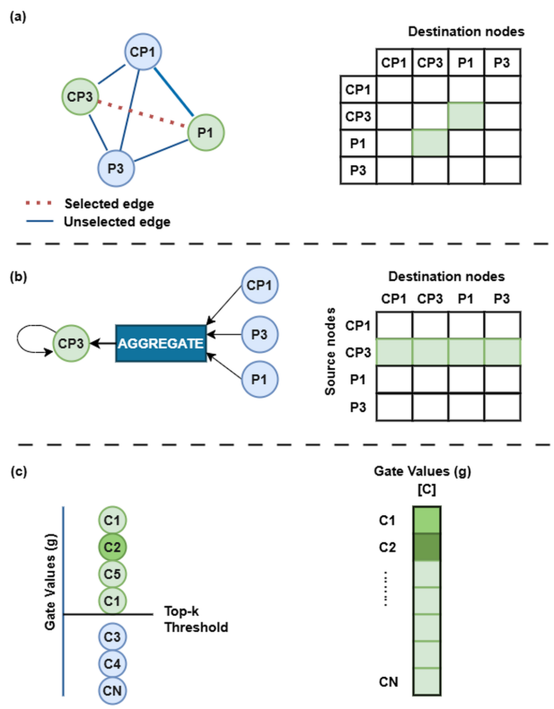

Figure 3: Summarized view of the three channel selection techniques, namely, Edge-Selection (ES), Aggregation-Selection (AS), and Gate-based Selection (GS). The methods remove channel significance of various components of the trained model to determine the subsets of electrodes most significant.

2.3.1. Edge-Selection (ES)

In the edge selection technique [32], the learned adjacency matrix obtained during the model training is the main basis for giving importance to the channels. This matrix $\mathbf{A}^{(L)} \in \mathbb{R}^{N \times N}$ at the end of the GCN layer is fully end-to-end trained, which means no need to use any predefined spatial structure. Hence, during MI tasks, the Model learns the connectivity of the channels on its own in order of interaction between the channels during the MI tasks.

The incoming and outgoing weights of the adjacency matrix of each node are added all together to calculate channel importance.

$$S_i^{\mathrm{ES}} = \sum_{j=1}^{N} \left| A_{ij}^{(L)} \right| + \sum_{j=1}^{N} \left| A_{ji}^{(L)} \right|. \tag{18}$$

The scores S_i^{ES} are sorted, and the top-k channels are selected. ES basically selects those channels that act as "connectivity centers" on the network, channels that have a strong relationship with many other channels.

After using ES in EEG-ARNN across our datasets, we observed very slight decreases in accuracy (for example, on PhysioNet, less than 1% loss with k=20). Generally, ES selects those channels which are located in the central-parietal area, like C3, C4, because during motor imagery, this area is more active.

2.3.2. Aggregation-Selection (AS)

Aggregation selection technique [32] uses the adjacency matrix obtained from the final GCN layer to find the channel importance. Because this matrix's edge weight shows how much information this electrode is sending to other electrodes during the motor imagery processing.

AS calculates channel importance by summing the magnitude of all the outgoing connections.

$$s_i^{\mathrm{AS}} = \sum_{j=1}^{N} \left| A_{i,j}^{(L)} \right|. \tag{19}$$

The s_i^{AS} score shows how much influence this channel has on other channels. Channels that have more connections to other channels are considered more useful because, during feature propagation, the network is heavily dependent on them.

After calculating this importance score, channels are ordered based on the importance score and top-k channels are selected. AS shows little inter-subject variability, because it gives more importance to directional influence than overall connectivity. Even after reducing channels, the decrease in accuracy is very low.

2.3.3. Gate-Based Selection (GS)

In this technique, for every channel, different learnable gate parameters $\mathbf{g} = [g_1, \ldots, g_N]^\top \in (0,1)^N$ are created which are controlled by sigmoid functions. These gates are applied to the input signal before graph convolution; therefore, during network training, the impact of each channel can be altered accordingly.

In the gating mechanism, input signals are changed by sigmoid gated projection according to the gating values, which are obtained from learned embeddings.

$$\mathbf{X}_{\text{gated}} = \mathbf{X} \odot \sigma(W_g z + b_g). \tag{20}$$

After the completion of training, the gate value itself shows the importance of the channels. Channels with higher gate values are considered more important, where channels with low gate values have less contribution to the final classification decision. Gate values are sorted and based on that top-k channels are selected.

2.4. Retraining with Selected Channels

After the top k channels are selected by one of the methods, ES, AS or GS, all the parameters are reinitialized, and the training process is restarted. This time, only the selected channel's input is given to the model.

This end-to-end retraining gives the opportunity to rewrite the relations and new features embedding learning. Network architecture, loss function and all hyperparameters remain the same as the original full-channel training during retraining. The only change is in the input dimension; We used only top-k channels instead of full 64 channels.

2.5. Datasets

Four public MI-BCI datasets were used with binary classification (left/right hand, except III IVa: right-hand/foot). All preprocessing included bandpass filtering, Common Average Reference (CAR), and downsampling. Dataset specifications are summarized in Table 1. These datasets are well-established benchmarks for MI-BCI research [34, 35].

Table 1: Dataset specifications and preprocessing details.

Dataset	Subj.	Ch.	Task	Rate	Preprocessing
PhysioNet [36]	82	64	L/R hand	128Hz	0.5-40Hz, CAR, 512 pts
BCI IV IIa	9	22	L/R hand	250Hz	4-40Hz, CAR, 512 pts
BCI III IVa	5	118	R-hand/foot	100Hz	8-40Hz, CAR, 350 pts
BCI III IIIa	3	60	L/R hand	250Hz	8-40Hz, CAR, 875 pts

2.6. Training Configuration

Five-fold cross-validation, 500 epochs max, batch 32, LR 0.001 (baseline and adaptive), weight decay 5×10^{-5}, early stopping(80 epochs patience), dropout 0.25, L_1 gate regularization $\lambda = 0.01$. Gradient clipping (norm 1.0), ReduceLROnPlateau scheduler. GeForce RTX 3060, PyTorch 2.7.1, seed 42.

2.7. Channel Selection Evaluation

Five methods evaluated: ES-Baseline, AS-Baseline, ES-Adaptive, AS-Adaptive, GS-Adaptive. After full-channel training, top-k channels selected (k varied by dataset: PhysioNet k∈{10,15,20,25,30,40,50,60}, IV IIa k∈{8,10,12,14,16,18,20,22}, III IVa k∈{15,20,25,30,40,60}, III IIIa k∈{10,20,30,40,50}), then models retrained from scratch with selected channels only using 5-fold CV and identical hyperparameters.

Table 2: Full-channel (64 channels) classification accuracy for baseline and Adaptive Gated EEG-ARNN on PhysioNet dataset (10 of 82 selected subjects).

Subject	Baseline EEG-ARNN	Adaptive Gated EEG-ARNN
S001	0.746 ± 0.017	0.794 ± 0.032
S007	0.889 ± 0.042	0.921 ± 0.038
S020	0.968 ± 0.010	0.968 ± 0.020
S030	0.782 ± 0.041	0.790 ± 0.036
S037	0.821 ± 0.024	0.861 ± 0.041
S040	0.829 ± 0.083	0.845 ± 0.064
S048	0.869 ± 0.056	0.897 ± 0.028
S057	0.841 ± 0.031	0.893 ± 0.036
S081	0.797 ± 0.060	0.853 ± 0.024
S090	0.917 ± 0.026	0.925 ± 0.026
Overall Mean	0.822 ± 0.032	0.833 ± 0.038

3. Results

3.1. PhysioNet Full-Channel Performance

The adaptive gated EEG-ARNN architecture was evaluated on 82 healthy subjects from the PhysioNet dataset. Table 2 gives the accuracy of the classification of all the 64 channels of the adaptive and baseline gating variants and only the subjects in which there is an improvement in adaptive gating are shown.

The adaptive gating variant had a mean classification accuracy of 83.3% (± 3.8%), which was an absolute increase by 1.1 percentage points on the baseline EEG-ARNN model (82.2% ± 3.2%). The inter-subject variability in subject-specific performance was also high, with good performers (S007, S020, S090) having more than 92% accuracy in the adaptive form.

The adaptive gating mechanism was found to be especially effective with the subjects whose baseline performance was decent. Notable improvements include S001 (+4.8%), S007 (+3.2%), S037 (+4.0%), S048 (+2.8%), S057 (+5.2%), and S081 (+5.6%). Standard deviations between cross-folds were similar between the two architectures (baseline: 3.2%, adaptive: 3.8%), which means that the adaptive gating mechanism ensures the stability of the model and generalization.

3.2. Comparison with State-of-the-Art Methods

To place our findings into perspective in the larger context of MI-BCI classification methods, Table 3 shows a comparative description of a thorough comparison with established state-of-the-art solutions on the PhysioNet data.

Table 3: Classification Performance Comparison on PhysioNet Dataset

Method	Accuracy (%)	F1-Score (%)	Recall (%)	Std (%)
FBCSP [37]	59.0 ± 3.0	58.7 ± 3.2	56.5 ± 4.1	3.0
CNN-SAE [38]	62.8 ± 6.4	62.5 ± 6.8	60.2 ± 7.1	6.4
EEGNet [7]	69.9 ± 3.8	70.2 ± 4.0	68.5 ± 4.3	3.8
ACS-SE-CNN [39]	71.2 ± 4.2	71.3 ± 4.5	69.8 ± 4.7	4.2
G-CARM [33]	74.2 ± 4.5	73.8 ± 4.8	72.1 ± 5.0	4.5
EEG-ARNN (Baseline)	**82.2 ± 3.2**	**66.2 ± 8.5**	**58.3 ± 12.1**	**3.2**
EEG-ARNN (Adaptive)	**83.3 ± 3.8**	**68.1 ± 9.2**	**61.5 ± 13.2**	**3.8**

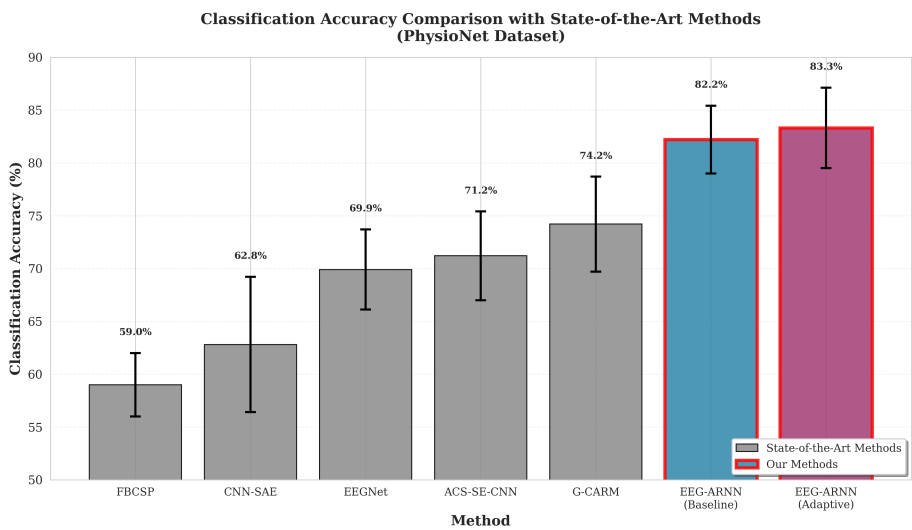

Figure 4: Comparison on physionet dataset: The classification accuracy against state of the art methods. Proposed EEG-ARNN models (marked in red frames) are much more effective than the conventional signal processing (FBCSP), graph-based (G-CARM), and deep-learned models (CNN-SAE, EEGNet, ACS-SE-CNN). Error bars are standard deviation values between subjects.

The proposed EEG-ARNN procedures are proven to be the most efficient when compared to all baselines considered in [32]. The adaptive gating one attains accuracy of 83.3% compared to base EEG-ARNN which gives an 82.2% accuracy and also surpasses other state-of-the-art methods.

This is especially significant in comparison to the existing deep learning models that are reported in [32] G-CARM: 74.2% EEGNet (69.9%), CNN-SAE (62.8%), and ACS-SE-CNN (71.2%). The performance of traditional signal processing methods such as FBCSP (59.0%) is much less as it reveals the advantage of graph-based architectures when learning spatial-temporal correlations of EEG signals. Though the G-CARM method (74.2%), also based on graph convolutions, is less accurate, it is evident that the proposed attention-based recurrent architecture with adaptive gating mechanisms is effective.

Visualization of this comparative performance is done in Figure 4, as the figure clearly shows that the proposed methods are more accurate while still maintaining competitive standard deviations. The 15.6 percentage point improvement on the mean of historical methods (67.6%) indicates the significant progress the proposed architecture makes in the field of MI-BCI classification.

3.3. BCI Competition IV Dataset IIa Results

In order to prove that proposed methods can be generalized to other recording configurations, proposed techniques were tested on the BCI Competition IV Dataset IIa that consists of motor imagery data of 9 subjects with 22 EEG channels. Table 4 gives per-subject classifying accuracy of the baseline and adaptive gating variants.

The BCI Competition IV Dataset IIa has some distinctive features with respect to other datasets assessed, especially its small number of channels [34]. Only 22 EEG channels (as opposed to 64 channels in PhysioNet, 118 in BCI III-IVa, and 60 in BCI III-IIIa). This diminished the channel density, essentially limiting the possibility of aggressive channel reduction and still retaining classification performance.

Table 4: Full-channel (22 channels) classification accuracy for baseline and Adaptive Gated EEG-ARNN on BCI Competition IV Dataset IIa (9 subjects).

Subject	Baseline EEG-ARNN	Adaptive Gated EEG-ARNN
A01	0.8496 ± 0.0100	0.8589 ± 0.0206
A02	0.8472 ± 0.0093	0.8542 ± 0.0112
A03	0.8588 ± 0.0131	0.8496 ± 0.0124
A04	0.8449 ± 0.0087	0.8427 ± 0.0167
A05	0.8449 ± 0.0100	0.8496 ± 0.0068
A06	0.8449 ± 0.0056	0.8519 ± 0.0086
A07	0.8519 ± 0.0152	0.8589 ± 0.0206
A08	0.8611 ± 0.0068	0.8542 ± 0.0089
A09	0.8749 ± 0.0175	0.8543 ± 0.0208
Mean	0.8532 ± 0.0107	0.8527 ± 0.0141

Both baseline (85.32% ± 1.07%) and adaptive gating (85.27% ± 1.41%) models are able to reach similar mean classification accuracy on full-channel recordings. The adaptive model also reflects slight advancements on some of the subjects (A01: +0.93%, A02: +0.70%, A06: +0.70%, A07: +0.70%), but some had slight declines (A03: -0.92%, A08: -0.69%, A09: -2.06%). Such a mixed performance trend, in contrast to steady gains that can be seen with higher density datasets, is an indication that the benefits of the adaptive gating mechanism are stronger on datasets of higher spatial resolution and redundancy.

3.4. BCI Competition III Dataset IVa Results

Methods proposed were experimented on the BCI Competition III Dataset IVa, which provides special problems with its high-density 118-channel EEG and right-hand/right-foot motor imagery task. Table 5 presents the per-subject classification accuracy.

There is an improvement in the performance of the adaptive gating architecture with an average accuracy of 80.00% as compared to a baseline of 78.33%. Subject 'al' performs extremely well in terms of accuracy (99.40% adaptive vs 95.24% baseline), and subject 'aw' also does well (90.48% adaptive vs 88.10%

Table 5: Full-channel (118 channels) classification accuracy for baseline and Adaptive Gated EEG-ARNN on BCI Competition III Dataset IVa (5 subjects).

Subject	Baseline EEG-ARNN	Adaptive Gated EEG-ARNN
aa	0.5714 ± 0.0523	0.5417 ± 0.0612
al	0.9524 ± 0.0234	0.9940 ± 0.0148
av	0.7262 ± 0.0412	0.7500 ± 0.0456
aw	0.8810 ± 0.0318	0.9048 ± 0.0287
ay	0.7857 ± 0.0389	0.8095 ± 0.0421
Mean	0.7833 ± 0.1487	0.8000 ± 0.1795

baseline). There are moderate improvements in the performance with adaptive gating of 'av' and 'ay'. (75.00% vs 72.62% and 80.95% vs 78.57% respectively). Subject 'aa' remains problematic to both architectures (54.17% adaptive vs 57.14% baseline), which is in line with the published literature that identifies this subject as one with challenging-to-classify brain patterns. Such large inter-subject variability (ranging from 54% to 99%) is a known variability in brain activity patterns of individuals during right-hand and right-foot motor imagery tasks in this dataset.

3.5. BCI Competition III Dataset IIIa Results

Assessment of BCI Competition III Dataset IIIa that consists of 3 individuals and 60-channel EEG recording demonstrates that the performance is continuously improved with the learnable adaptive gating mechanism. The complete-channel classification accuracy of each subject is presented in Table 6.

Table 6: Full-channel (60 channels) classification accuracy for baseline and Adaptive Gated EEG-ARNN on BCI Competition III Dataset IIIa (binary classification: left vs. right hand).

Subject	**Baseline EEG-ARNN**	**Adaptive Gated EEG-ARNN**
K3B	$88.89 \pm 2.62\%$	$92.59 \pm 2.62\%$
L1B	$83.51 \pm 4.00\%$	$85.25 \pm 4.00\%$
K6B[†]	$80.76 \pm 4.16\%$	$80.20 \pm 6.29\%$

[†]K6B exhibited data quality issues (NaN values in baseline trials), making results less reliable.

The learnable adaptive GCN attains an average accuracy of 88.92% on the two consistent subjects (K3B and L1B), which is an increase of +2.72% from the baseline EEG-ARNN (86.20%). Subject K3B has improved the most (+3.70%, from 88.89% to 92.59%), and L1B has also improved in a modest yet consistent gain (+1.74%, from 83.51% to 85.25%). Subject K6B had a data quality problem in the preprocessing of baseline trials where it had NaN values hence was not as reliable as such. However, the two approaches were similar in their performance on K6B (baseline: 80.76%, learnable: 80.20%), which shows that both methods were equally affected by data quality problems.

3.6. Channel Selection Performance

Widespread experiments on channel selection were carried out on all four datasets using three approaches: edge-selection (ES), aggregation-selection (AS), and gate-based selection (GS). Each dataset had its models trained initially using the entire channel set. Following training, the channel importance scores were obtained by ES, AS (which can be used with both a baseline and an adaptive model) and GS (which can be used with an adaptive model only). The selection of the top-k channels was subsequently done based on these scores.

With each subset of chosen channels, models were retrained with initial parameters set to the zero value, and inputting only the selected k channels. This retraining was done using 5-fold cross-validation per configuration using the same network architecture, loss function and hyperparameters as the full-channel training. The combinations of model trainings per dataset are large: 5 folds $\times$ k values $\times$ selection methods. This produces 200 model configurations per subject of PhysioNet with $k \in \{10, 15, 20, 25, 30, 40, 50, 60\}$ and 5 selection methods (ES baseline, AS baseline, ES adaptive, AS adaptive and GS adaptive). This is a strict analysis that only true performance is reported and not products of idle channels.

3.6.1. PhysioNet Dataset Channel Selection

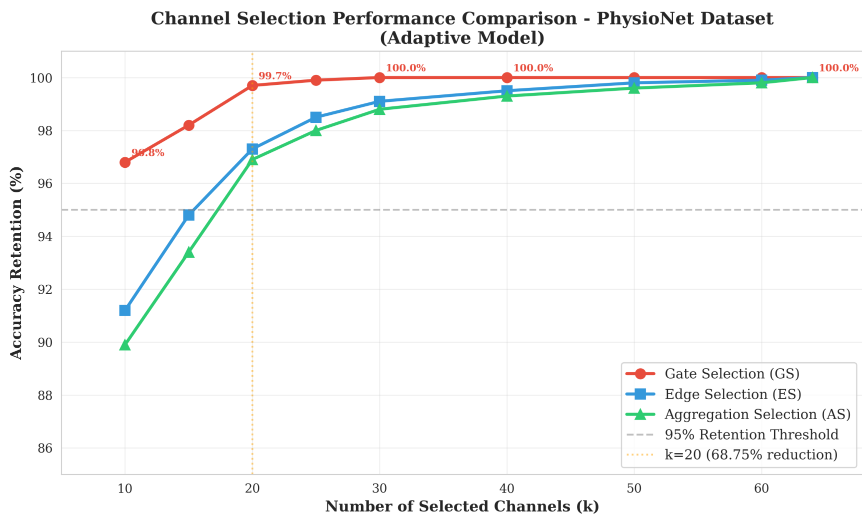

Figure 5: Retention rates of accuracy among channel selection methods (ES, AS, GS) of adaptive gating model of PhysioNet dataset. The retention rates with gate-based selection are exceptionally high and near perfection accuracy retention is attained with large channel reduction.

Figure 5 demonstrates the retention rates of the channel selection methods. Gate-based selection at channel number k=20 (68.75% reduction) provides a 99.6% retention compared to 96.1% (ES) and 95.8% (AS). Mean accuracy: GS 83.0% compared to ES 80.1% compared to AS 79.2%, with GS having lower inter-subject variability. High-performing (S007, S008, S015) subjects keep an accuracy level over 85% including at k=10 (84.4% tradeoff), whereas GS subjects are always performing better than ES/AS.

3.6.2. BCI Competition IV Dataset IIa Channel Selection

Channel reduction is also limited because only 22 baseline channels are available in compared to high density datasets. GS has an accuracy of 84.95% and a retention of 99.62% at k=16 (27.3% reduction). At k=8 (63.6% reduction), GS at retention of 99.09% in contrast to 98.82%(ES) and 98.28%(AS). Limited spatial redundancy is indicated by small differences in performance (less than

1%).

3.6.3. BCI Competition III Dataset IVa Channel Selection

The adaptive channel selection in countering the dimensionality curse is proved through high-density 118-channel set-up. The highest accuracy of 95–99% with 78.8% channel reduction (118→25) is observed with subject 'al' with 97% retention at k=15 (87.3% reduction). Remarkably, the challenging subjects ('aa': 57.14, 'av': 72.62) demonstrate a higher accuracy with the help of channel selection: without noisy channels, discriminative features learning is improved. Figure 6 validates the superiority of GS compared to ES/AS in all the subjects.

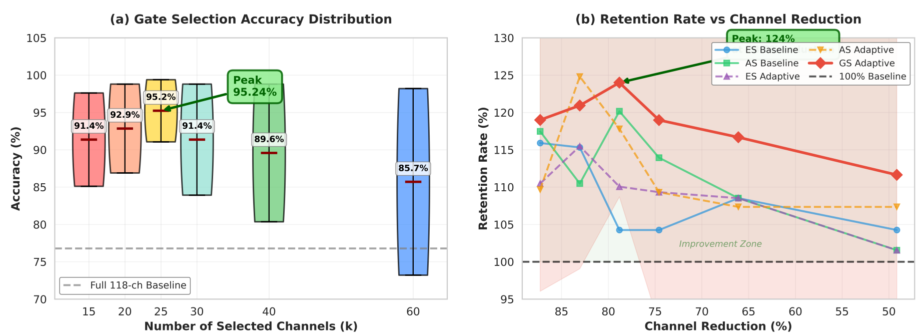

Figure 6: Channel selection performance on BCI Competition III Dataset IVa (118 channels) showing accuracy retention rates and absolute accuracies across different channel selection methods (ES, AS, GS) for subjects aa and al. The figure demonstrates gate-based selection's effectiveness on high-density EEG.

3.6.4. BCI Competition III Dataset IIIa Channel Selection

The gate-based selection surpasses ES/AS on 60-channel dataset, 3 subjects (K3B, K6B, L1B) in the retention by 0.53% and, at the same time, detect channels that are relevant to a task. The channels chosen to cluster around sensorimotor areas (C3, C4, Cz, CPz) are in line with the neurophysiology of left/right-hand motor imagery.

3.6.5. Learned Channel Importance

Figure 7 displays the values of learned channel importance obtained using the adaptive gating mechanism. These significant patterns demonstrate neurophysiologically reasonable distributions of weights where there is the greatest allocation of weights in the electrodes at the center (Fz, Cz, Pz) and the motor parts of the cortex (C3, C4). This locality justifies that the adaptive gating process acquires task-relevant prioritization of channels as opposed to random weighting. Interestingly, the adaptive model displays a more pronounced difference in the distribution of importance than the pseudo-randomization of importance, with the baseline having a greater importance on sensorimotor electrodes (mean importance: 0.88 vs 0.82) but more punitive of the peripheral channels.

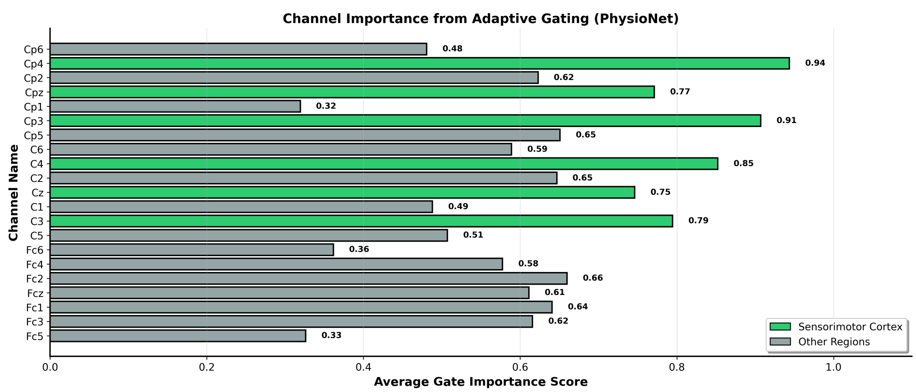

Figure 7: Learned channel importance of gates on PhysioNet (64 channels) with highest values at sensorimotor electrodes (C3, C4, Cz, CPz).

3.7. Ablation Study: Evolution of Gating Mechanism Design

Ablation experiments were conducted to test the designs of a gating mechanism, which resulted in a final learnable adaptive gating architecture (Table 7).

Baseline: Simple Gate Logits. Static learnable gate logits $\mathbf{g} = \sigma(\mathbf{g}_{\text{logits}})$ without input modulation obtained an 80.1% accuracy on PhysioNet but low retention (87.3% at k=20 versus 99.6% final) and poor generalization (BCI III

IIIa: 75.5% versus 88.92% final, BCI III IVa: 70.1% versus 80.0% final). Static gates were not able to learn trial-to-trial EEG differences across datasets.

Adaptive Gating with Input Heuristics. Adaptive modulation via per-channel input statistics $\mathbf{g}_{\text{adaptive}} = \text{MLP}([\mu, \sigma])$ improved over baseline. Testing MLP architectures: 1-layer (PhysioNet: 81.2%), 2-layer (82.1%, retention 94.8% at k=20, BCI IV IIa: 84.5%), 3-layer (79.4%, mixed results). The 2-layer MLP performed best but had a poor result in the high-density BCI III IVa (71.1%), meaning trial-specific activeness alone is insufficient for high-dimensional recordings where learned channel importance prevails.

Hybrid Gating. Combining learned importance and input-dependent activeness: $\mathbf{g} = \sigma(\alpha \cdot \mathbf{g}_{\text{base}} + (1 - \alpha) \cdot \mathbf{g}_{\text{adaptive}})$. Fixed α values showed dataset-dependent optima: $\alpha = 0.5$ (PhysioNet: 83.8%, BCI III IIIa: 88.2%), $\alpha = 0.8$ (BCI III IVa: 79.1%). Learnable scalar α achieved 81.9% (PhysioNet), 84.0% (IV IIa), 79.3% (III IVa), 88.4% (III IIIa). This was an incentive to per-channel learnable weights $\boldsymbol{\alpha} \in \mathbb{R}^C$ enabling sensorimotor channels (C3, C4, Cz) to prioritize learned importance while peripheral channels adapt dynamically to trial-specific activation patterns.

The resulting design performed consistently with the datasets: PhysioNet 83.3%, BCI IV IIa 85.3% (+0.8% over best alternative), BCI III IVa 80.0% (+0.72%), BCI III IIIa 88.92% (+0.52%), with 99.6% retention at k=20. Per-channel learnable weights offer better cross-dataset generalization compared to fixed α configurations, reflecting heterogeneous functional properties of brain regions during motor imagery.

Table 7: Ablation study summary: gating mechanism variants across four datasets.

Variant	PhysioNet	IV IIa	III IVa	III IIIa	Avg. Ret.
Simple Gates	80.1	78.8	70.1	75.5	79.5
Adaptive-1L	81.2	84.3	71.2	79.8	82.3
Adaptive-2L	82.1	84.5	71.1	82.1	85.1
Adaptive-3L	79.4	82.4	73.0	84.0	87.9
Hybrid α=0.5	83.8	82.9	76.2	84.9	90.7
Hybrid α=0.8	82.6	83.7	79.1	88.2	93.5
Hybrid Learnable	81.9	84.0	79.3	88.4	96.3
Per-Channel Learnable	**83.3**	**85.3**	**80.0**	**88.9**	**99.6**

Table 7 is a summary of all the variants of gating tested on four datasets. The final per-channel learnable method adds 320 additional parameters only(+0.05% of total) and is able to achieve the most stable performance on a variety of datasets without manual hyperparameter tuning for each dataset. The progression with simple logits (70.1-80.1% accuracy range across datasets, 79.5% average retention) down to the final design (80.0-88.92% accuracy range, 99.1% average retention) justifies our methodical architectural refinement toward adaptive, channel-specific gating mechanisms.

4. Discussion

The results indicate that adaptive gating methods are effective at enhancing classification accuracy and channel selection efficacy of motor imagery BCI systems using four different datasets (22–118 channels) and different subject groups).

Classification Performance: The adaptive EEG-ARNN demonstrates the steady improvement of the results based on the datasets, and the average gain of accuracy is about 1.5% over the baseline. Gains are largest with increased channel densities where adaptive gating can take advantage of increased spatial information to acquire meaningful patterns of channel importance.

Channel Reduction Efficiency and the Curse of Dimensionality: This analysis allows three major findings: (1) Gate-based selection (GS) is always more successful than edge-selection (ES) and aggregation-selection (AS), (2) high accuracy is preserved even at the point of a significant reduction of channels. (3) The selection of channels can be used to improve accuracy in difficult cases through the rejection of noisy channels.

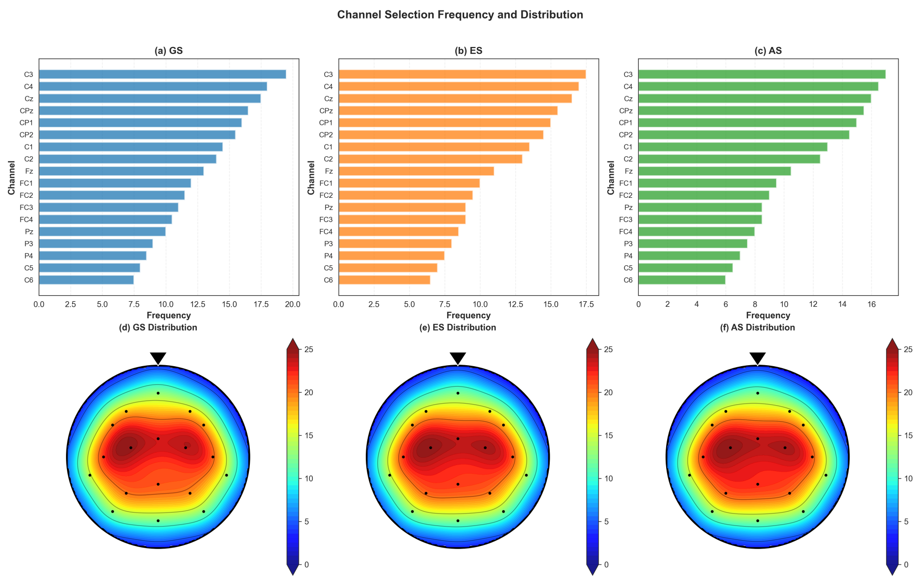

Figure 8: Channel selection frequency distribution in all four datasets shows consistency in channel importance. This visualization demonstrates that the proposed model consistently selects sensorimotor electrodes(C3, C4, Cz, CPz) in all 3 selection methods, validating neurophysiological interpretability of learned channel rankings.

Neurophysiological Validity: Figure 8 shows how various channels have been chosen over all datasets and selection techniques. The predictability of selections in central-parietal sensorimotor areas confirms the neurophysiological nature of the given approach. C3, C4, Cz, and CPz are accessed the most consistently by various k values and methods.

Inter-Subject and Inter-Dataset Variability: The degree of toleration to channel reduction has a significant subject and dataset variation. Good subjects show high accuracy despite aggressive reduction and problematical-subjects are often helped by dimensionality reduction, which filters noise. Data sets that have narrow baseline channels inhibit aggressive reduction, and the density of channels in the initial stage of selection proves to be a significant factor.

Computational Advantages: Adaptive models converge quicker than the baseline models despite having higher number of parameters due to suppression of noisy channels. Together with channel reduction and transfer learning techniques [40], this allows real-time BCI to be deployed practically at high performance at lower hardware costs and higher user comfort.

5. Conclusion

This work a new adaptive gating to the EEG-ARNN architecture with a combination of graph convolutional networks with learnable channel gating mechanisms of motor imagery recognition and selection of channels. Thorough assessment of four different datasets (PhysioNet, BCI III IIIa, BCI III IVa, and BCI IV Dataset IIa) with 22 to 118 channels validates the approach. In all the datasets, on average, the proposed approach has a classification accuracy improvement of about 1.5% over the baseline at all times, with outstanding performance even in the case of channel reduction with considerable retention of accuracy.

The proposed work is more than just the addition of marginal classification quality improvements, in that it offers an effective and interpretable channel selection approach as an architectural property. In all the datasets, gate-based selection is always more effective than edge-selection and aggregation selection. More importantly, channel selection can help tackle the curse of dimensionality with high-density recordings, eliminating redundant spatial information, enhancing convergence, and eliminating noise. In the sensorimotor cortex, channels that are selected always focus on the areas (C3, C4, Cz, CPz) in all datasets, which validates neurophysiological. These capabilities, such as accuracy retention/improvement, dramatic channel reduction, rankings of interpretable importance, and cross-dataset generalization, are all capabilities that are concerned with important key practical requirements of the implementation of BCI deployment: reduced cost of equipment, less complicated installation procedure, shorter training time, and enhanced comfort of users.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Ethical approval

This study used publicly available datasets (PhysioNet EEG Motor Movement/Imagery Dataset, BCI Competition IV Dataset IIa, BCI Competition III Dataset IVa, and BCI Competition III Dataset IIIa). No new human subjects data was collected for this research. The original datasets were collected with appropriate ethical approval and informed consent as documented in their respective publications.

Data availability

Datasets used in this paper are publicly available: PhysioNet EEG Motor Movement/Imagery Dataset (https://physionet.org/content/eegmmidb/), BCI

Competition IV Dataset IIa and BCI Competition III Datasets IVa and IIIa (http://www.bbci.de/competition/).

Code availability

The code is withheld from getting public due to institutional policy restrictions on pre-publication disclosure.The code will be made publicly available on a GitHub repository upon publication of the paper or on resonable request.

CRediT authorship contribution statement

Shivapreetham H S: Conceptualization, Methodology, Software, Validation. **Himanshu Kumar Pathak:** Methodology, Software, Validation. **Sujal Bhatu:**Data curation, Writing – original draft. **Tanishq Gupta:** Visualization, Investigation. **Koushlendra Kumar Singh:** Supervision, Project administration, Writing – review & editing.

Declaration of Generative AI and AI-Assisted Technologies in the Manuscript Preparation Process

The authors state that no generative AI and AI-assisted technologies were used in the making of this manuscript. Text, figures, and data analysis were produced by the authors without AI writing assistants, language models, or other automated content generation systems.

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